

Electoral Turnover and Forest Cover in Brazil: Close-Elections RD Estimates for Three Forest Outcomes

Generated by `run_rd_brasil.py`

June 14, 2026

Contents

1	Data and outcome construction	2
1.1	Pipeline	2
1.2	Outcomes ($K = 3$)	2
1.3	Marx–Pons–Rollet ΔY construction	2
1.4	The RD design	2
2	RD results	3
2.1	Main results table	3
2.2	RD plots (main horizon)	4
2.3	McCrary density tests	5

1 Data and outcome construction

1.1 Pipeline

Three production stages: (i) a Python data builder (`B_codes/build_rd_panel_brasil.py`) that consumes TSE election data (via BigQuery / Base dos Dados), MapBiomas Collection 10 forest layers, and MODIS VCF tree cover, and writes `rd_panel_turnover_bosque.csv` (one row per municipality-election); (ii) this analysis script (`run_rd_brasil.py`) that constructs the five MPR horizons, estimates the RD via `rdrobust`, runs the McCrary density test via `rddensity`, and generates `rdplots`; (iii) this report, assembled automatically by the same script.

1.2 Outcomes ($K = 3$)

Three forest-cover measures, each averaged at the municipality level:

- **native_mb** — MapBiomas Collection 10 native forest (`forest_native_ha`), annual area in hectares, 2000–2024.
- **formation_mb** — MapBiomas Collection 10 forest formation (`forest_formation_ha`), annual area in hectares, 2000–2024.
- **treecover_modis** — MODIS VCF (MOD44B v061) `Percent_Tree_Cover`, municipality-level mean, 2000–2024.

1.3 Marx–Pons–Rollet ΔY construction

For each (municipality m , election year t),

$$\Delta Y_{m,t}^{\text{main}} = \frac{1}{4} \sum_{\tau=1}^4 Y_{m,t+\tau} - Y_{m,t-1},$$

the post-election four-year mean minus the pre-election value. Shorter horizons replace the $1/4$ post-mean by $1/k$ for $k \in \{1, 2, 3\}$; the `pre3` robustness spec replaces $Y_{m,t-1}$ with the mean of $Y_{m,t-3..t-1}$. All ΔY are z -standardised using the (μ, σ) from the full main-horizon sample.

1.4 The RD design

- **Running variable.** $\text{score} = \text{margin} \times \widetilde{\text{turnover}}$, where $\text{margin} = \text{vote share of the winner} - \text{vote share of the runner-up}$, and $\widetilde{\text{turnover}} = +1$ if the winning party changed (turnover) and -1 if the same party re-won (continuity).
- **Treatment at the cutoff.** Party-level mayoral turnover in a close election. Positive score $\rightarrow$ turnover; negative $\rightarrow$ continuity.
- **Sample.** Brazilian municipal elections 2000, 2004, 2008, 2012, 2016, 2020, 2024 (5,568 municipalities, 32,453 muni-election cells with valid running variable). Average turnover rate: $\bar{T} = 0.725$.
- **Estimator.** `rdrobust` with $p = 1$, triangular kernel, MSE-optimal bandwidth (`bwselect("mserd")`). Reported: bias-corrected point estimate with robust SE and p -value (“Robust” line).

2 RD results

2.1 Main results table

Table 1: RD estimates: turnover $\rightarrow$ forest cover (Brazil). Per-cell: bias-corrected coefficient (top), robust SE (parentheses), MSE-optimal bandwidth h , and effective number of observations N_h . $p = 1$, triangular kernel.

Horizon	Native forest (MB)			Forest form. (MB)			Tree cover (MODIS)		
	Coef.	h	N_h	Coef.	h	N_h	Coef.	h	N_h
ΔY^{main}	-0.061** (0.031)	13.99	16,047	-0.036 (0.030)	14.34	16,333	-0.070* (0.042)	14.45	16,413
$\Delta Y^{k=1}$	-0.055* (0.031)	13.22	15,474	-0.028 (0.030)	13.60	15,744	-0.025 (0.041)	16.20	17,665
$\Delta Y^{k=2}$	-0.055* (0.031)	13.25	15,489	-0.030 (0.030)	13.45	15,628	-0.038 (0.041)	15.47	17,164
$\Delta Y^{k=3}$	-0.058* (0.031)	13.75	15,865	-0.033 (0.030)	13.84	15,941	-0.063 (0.043)	14.49	16,444
ΔY^{pre_3}	-0.061** (0.030)	14.83	16,715	-0.036 (0.030)	14.93	16,784	-0.043 (0.041)	16.18	17,657
McCrary p_q	0.000†			0.000†			0.000†		

Notes. Stars on the RD coefficient: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (robust p -value). † Marker on McCrary $p_q < 0.10$: weak evidence of running-variable manipulation at the cutoff.

2.2 RD plots (main horizon)

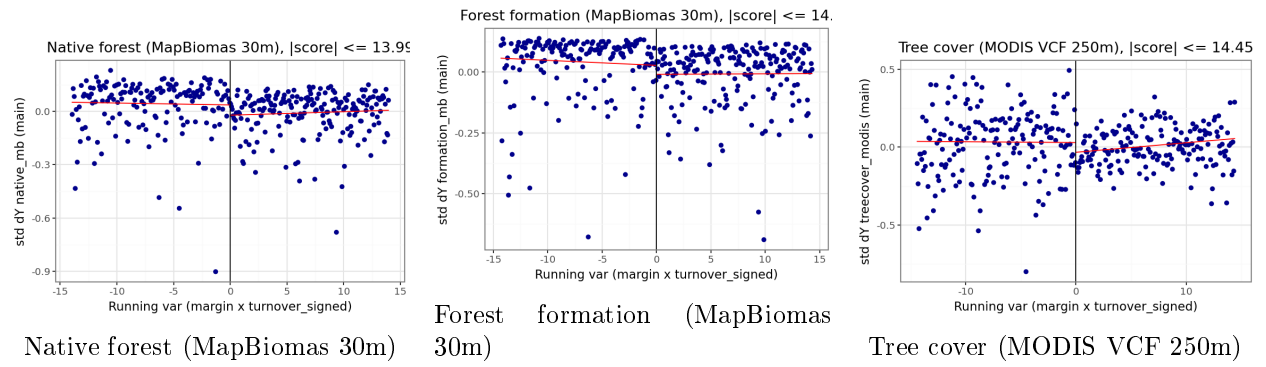


Figure 1: RD plots for the three forest outcomes (main horizon), restricted to the MSE-optimal bandwidth. Score: $\text{margin} \times \text{turnover}$, binselect = esmv.

2.3 McCrary density tests

Table 2: McCrary (rddensity) test on the running variable at the cutoff ($c = 0$). H_0 : no manipulation. † flags $p_q < 0.10$.

Outcome	T_q	p_q	N_l	N_r
<code>native_mb</code>	18.416	0.000†	7,217	19,790
<code>formation_mb</code>	18.416	0.000†	7,217	19,790
<code>treecover_modis</code>	18.416	0.000†	7,217	19,790